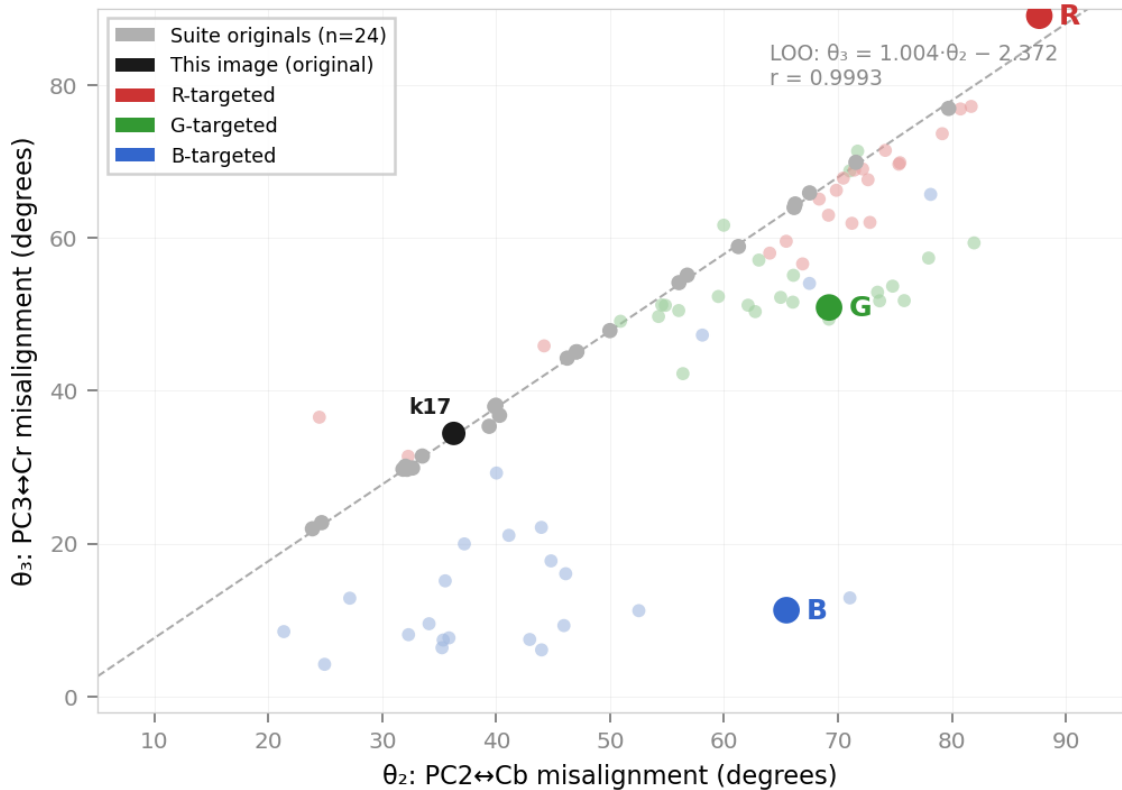


Compression Profile: KODIM17512x768 | 24-bit RGB | PNG (lossless)

monument | Near-Degenerate (PC1 = 97.78%)

1. Geometric Displacement Under Directional Perturbation



2. Pairwise Correlation and Angular Misalignment

Version	R-G	R-B	G-B	Avg r	theta2	theta3	L00 dev
Original	0.9852	0.9334	0.9777	0.965	36.27°	34.41°	+0.37°
R-targeted	0.0585	0.2047	0.9704	0.411	87.68°	89.04°	+3.38°
G-targeted	-0.1060	0.9681	-0.1012	0.392	69.22°	50.88°	-16.24°
B-targeted	0.9674	-0.2041	-0.1327	0.435	65.48°	11.29°	-52.08°

R-targeted isolates red (R-G, R-B < 0.20) while preserving G-B coupling at 0.970.
G-targeted isolates green (R-G, G-B < 0.11) while preserving R-B at 0.968.
B-targeted preserves R-G at 0.967 while driving R-B and G-B near zero.
All three perturbation directions depart from the L00 constraint manifold ($r = 0.9993$).

Compression Profile: KODIM17

512x768 | 24-bit

RGB | PNG (lossless)

monument | Near-Degenerate (PC1 = 97.78%)

3. File Format and Size Comparison

All versions depict the same scene. BPP = (file size in bits) / (pixel count).

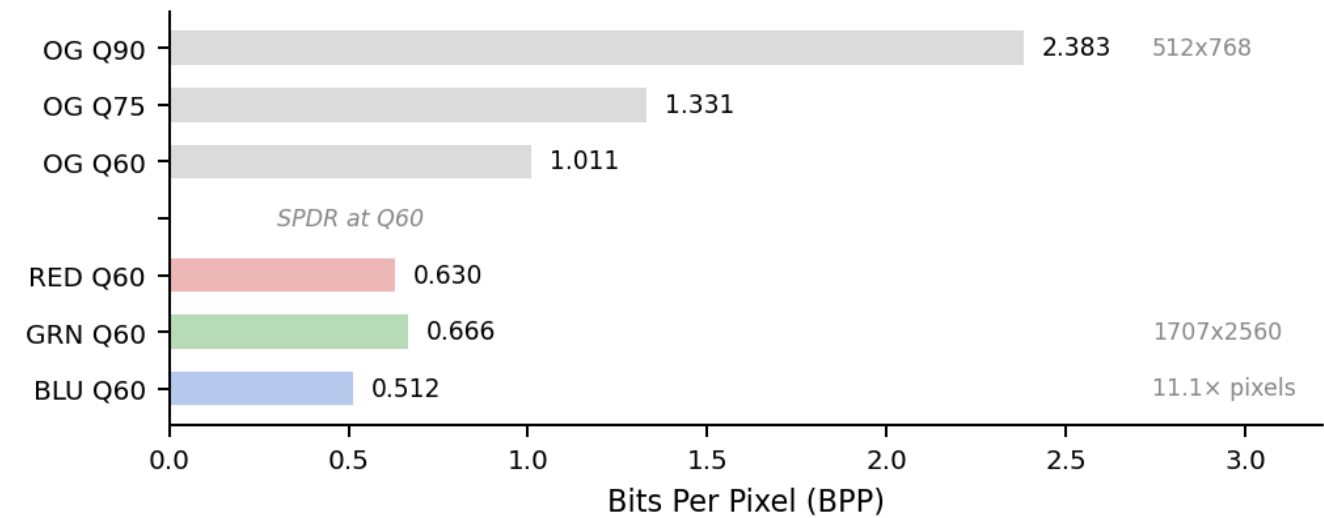
ORIGINAL (kodim17)

Version	Format	Resolution	Size	BPP	Pixels
PNG (lossless)	PNG	512x768	588.0 KB	12.249	393,216
JPEG Q90	JPEG	512x768	114.4 KB	2.383	393,216
JPEG Q75	JPEG	512x768	63.9 KB	1.331	393,216
JPEG Q60	JPEG	512x768	48.5 KB	1.011	393,216

REDISTRIBUTED (kodim17 SPDR at Q60)

Version	Format	Resolution	Size	BPP	Pixels
RED Q60	JPEG	1707x2560	336.3 KB	0.630	4,369,920
GRN Q60	JPEG	1707x2560	355.0 KB	0.666	4,369,920
BLU Q60	JPEG	1707x2560	273.1 KB	0.512	4,369,920

4. Quality Ladder: SPDR Q60 vs Original Q90



All three perturbation directions at Q60 produced lower BPP than the original at Q90, while carrying 11.1x the original pixel count.

Compression Profile: KODIM17 512x768 | 24-bit RGB | PNG (lossless)

monument | Near-Degenerate (PC1 = 97.78%)

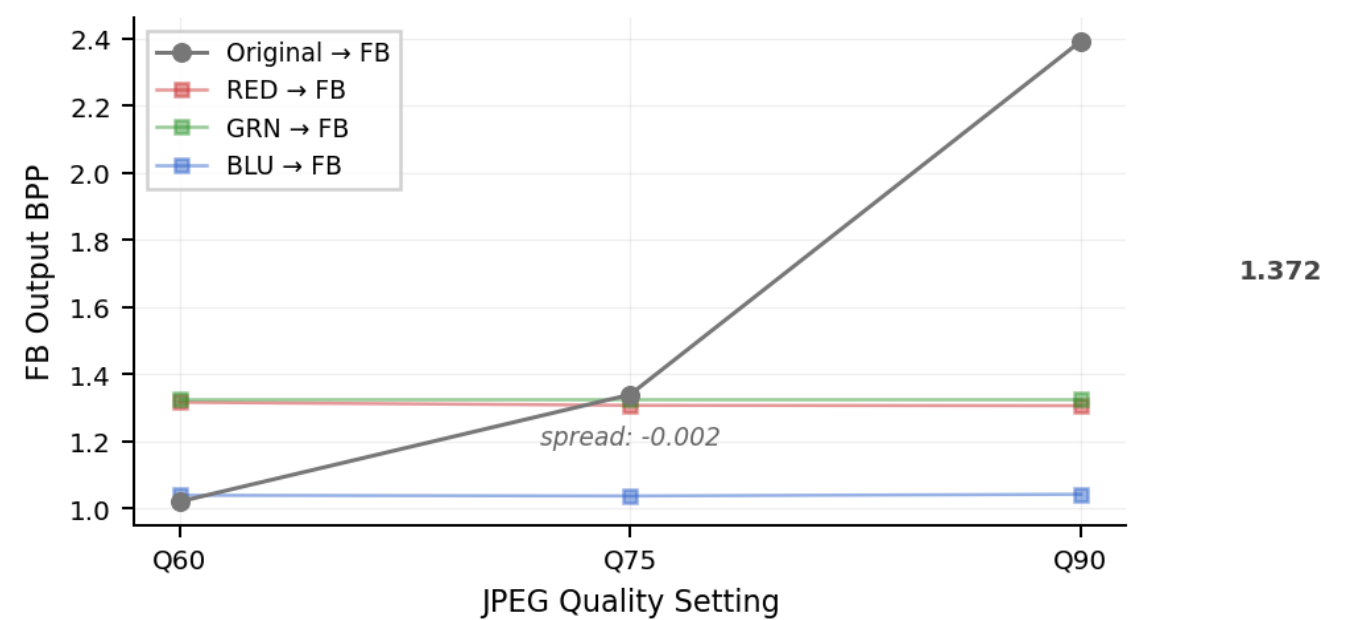
5. Third-Party Compression Pipeline (Facebook/Meta)

Each version uploaded to Facebook. Platform-compressed output measured at steady state (FB2).

Pipeline	Input Res	Output Res	FB BPP	BPP reduction
OG PNG → FB	512x768	512x768	2.249	—
R-targeted → FB	1707x2560	1366x2048	1.307	41.9%
G-targeted → FB	1707x2560	1366x2048	1.327	41.0%
B-targeted → FB	1707x2560	1366x2048	1.043	53.6%
ABC (no perturbation) → FB	1707x2560	1366x2048	1.227	45.4%

ABC control: same pipeline, same resolution, no covariance perturbation. Avg |r| unchanged (0.9659 vs 0.965 original). Comparable BPP reduction was not observed in the ABC control.

6. Quality Parameter Sensitivity Through Facebook



Original: Q60 → Q90 produces a 1.372 BPP spread through Facebook.
Redistributed: Q60 → Q90 produces a 0.002 BPP spread through Facebook.
Redistributed versions produced substantially smaller Q60-90 BPP variation through the measured pipeline.

7. Pipeline Summary – kodim17

Best BPP (Q60, direct)	0.512	1707x2560, 4,369,920 px
Best BPP through FB	1.043	1366x2048 output
vs OG Q90 → FB	56.4%	SPDR Q60→FB vs OG Q90→FB
Pixel ratio	11.1×	4,369,920 vs 393,216